Customer Shopping Behavior Analytics

End-to-End Data Pipeline: Python Preprocessing • MySQL Hypothesis Testing • Power BI Dashboard

Dataset: 3,900 Records | Stack: Python (Pandas/SQLAlchemy), MySQL, Power BI | Author: Umair Asif

Executive Summary

This project demonstrates an enterprise-grade analytics pipeline executed across 3,900 customer shopping behavior transactions. The workflow standardizes raw behavioral records in Python, executes relational business intelligence queries across 10 critical operational questions in MySQL, and packages insights into an interactive executive Power BI dashboard for data-driven customer acquisition, loyalty monetization, and margin optimization.

1. Python Data Cleaning & Feature Engineering Pipeline

The raw dataset (`customer_shopping_behavior.csv`) was imported and cleaned in Jupyter Notebook using Pandas and SQLAlchemy:

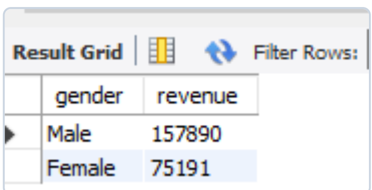
- **Categorical Median Imputation:** Identified 37 missing records in `Review Rating`. Preserved category distributions by imputing values with category-specific medians via `df.groupby('Category')['Review Rating'].transform(lambda x: x.fillna(x.median()))`.
- **Schema Normalization:** Converted all column headers to lowercase snake_case (e.g., `purchase_amount`) for seamless SQL schema mapping.
- **Multicollinearity Elimination:** Verified 100% equivalence between `discount_applied` and `promo_code_used` (`(df['discount_applied'] == df['promo_code_used']).all() == True`) and dropped the redundant promo code feature.
- **Age Quantile Binning (age_gab):** Discretized customer ages into four balanced quartiles (`Young Adult`, `Adult`, `Middle-aged`, `Senior`) using `pd.qcut()`.
- **Frequency Interval Mapping (`purchase_frequency_day`):** Mapped qualitative shopping cadences into numeric days (Weekly: 7, Fortnightly/Bi-Weekly: 14, Monthly: 30, Quarterly/Every 3 Months: 90, Annually: 365).
- **Automated Warehouse Ingestion:** Automated data streaming directly to MySQL (`customer_behavior.customer_data`) using SQLAlchemy's `create_engine`.

2. SQL Business Analysis & Verified Query Results

Q1: Total Revenue Generated by Male vs. Female

```sql
SELECT gender, SUM(purchase_amount) AS revenue
FROM customer_behavior.customer_data
GROUP BY gender;
```

Key Finding: Male customers generated **$157,890** (67.7%) compared to **$75,191** (32.3%) from Female customers.


Result Grid | Filter Rows:

gender	revenue
Male	157890
Female	75191

Q2: Discount Users Spending More Than Average Purchase Amount

```
SELECT customer_id, purchase_amount
FROM customer_behavior.customer_data
WHERE discount_applied = 'Yes'
  AND purchase_amount >= (
    SELECT AVG(purchase_amount)
    FROM customer_behavior.customer_data
  );
```

Key Finding: Identified high-value customers spending between **$64 and $97** despite receiving promotional discounts.

Result Grid | Filter Rows:

customer_id	purchase_amount
2	64
3	73
4	90
7	85
9	97
12	68
13	72
16	81
20	90

Q3: Top 5 Products with the Highest Average Review Rating

```
SELECT item_purchased,
       ROUND(AVG(review_rating), 2) AS "average review
rating"
FROM customer_behavior.customer_data
GROUP BY item_purchased
ORDER BY review_rating DESC
LIMIT 5;
```

Key Finding: Top-rated items are **Gloves (3.86)**, **Sandals (3.84)**, **Boots (3.82)**, **Hat (3.80)**, and **Skirt (3.78)**.

Result Grid | Filter Rows:

item_purchased	average_review_rating
Gloves	3.86
Sandals	3.84
Boots	3.82
Hat	3.8
Skirt	3.78

Q4: Average Purchase Amounts: Standard vs. Express Shipping

```
SELECT shipping_type,
       ROUND(AVG(purchase_amount), 2)
FROM customer_behavior.customer_data
WHERE shipping_type IN ('Standard', 'Express')
GROUP BY shipping_type;
```

Key Finding: Express shipping averages **$60.48** per transaction compared to **$58.46** for Standard shipping.

Result Grid | Filter Rows:

shipping_type	ROUND(avg(purchase_amount),2)
Express	60.48
Standard	58.46

Q5: Spend & Total Revenue: Subscribed vs. Non-Subscribed

```
SELECT subscription_status,
       COUNT(customer_id) AS
total_customers,

ROUND(AVG(purchase_amount), 2)
AS average_spend,

ROUND(SUM(purchase_amount), 2)
AS total_revenue
FROM
customer_behavior.customer_data
GROUP BY subscription_status;
```

subscription_status	total_customers	average_spend	total_revenue
Yes	1053	59.49	62645
No	2847	59.87	170436

> **Key Finding:** Non-subscribers generated **$170,436** across 2,847 shoppers ($59.87 AOV); subscribers generated **$62,645** across 1,053 shoppers ($59.49 AOV).

Q6: Top 5 Products with Highest Discount Utilization Rate

```
SELECT item_purchased,
       ROUND(SUM(CASE WHEN discount_applied = 'Yes'
THEN 1 ELSE 0 END)/COUNT(*)*100, 2) AS discount_rate
FROM customer_behavior.customer_data
GROUP BY item_purchased
ORDER BY discount_rate DESC
LIMIT 5;
```

item_purchased	discount_rate
Hat	50.00
Sneakers	49.66
Coat	49.07
Sweater	48.17
Pants	47.37

> **Key Finding:** High discount dependency observed on **Hat (50.00%)**, **Sneakers (49.66%)**, **Coat (49.07%)**, **Sweater (48.17%)**, and **Pants (47.37%)**.

Q7: Customer Segmentation (New, Returning, Loyal)

```
WITH customer_type AS (
  SELECT customer_id, previous_purchases,
    CASE
      WHEN previous_purchases = 1 THEN 'New'
      WHEN previous_purchases BETWEEN 2 AND 10
THEN 'Returning'
      ELSE 'Loyal'
    END AS customer_segment
  FROM customer_behavior.customer_data
)
SELECT customer_segment, COUNT(*) AS
number_of_customers
FROM customer_type
GROUP BY customer_segment;
```

customer_segment	number_of_customers
Loyal	3116
Returning	701
New	83

Key Finding: Customer distribution is heavily retention-driven with **3,116 Loyal** (>10 purchases), **701 Returning** (2–10), and **83 New** (1 purchase).

Q8: Top 3 Most Purchased Products Within Each Category

```
WITH item_count AS (
  SELECT category, item_purchased, COUNT(customer_id)
AS total_orders,
    ROW_NUMBER() OVER(PARTITION BY category ORDER BY
COUNT(customer_id) DESC) AS item_rank
  FROM customer_behavior.customer_data
  GROUP BY category, item_purchased
)
SELECT item_rank, category, item_purchased,
total_orders
FROM item_count
WHERE item_rank <= 3;
```

item_rank	category	item_purchased	total_orders
1	Accessories	Jewelry	171
2	Accessories	Sunglasses	161
3	Accessories	Belt	161
1	Clothing	Blouse	171
2	Clothing	Pants	171
3	Clothing	Shirt	169
1	Footwear	Sandals	160
2	Footwear	Shoes	150
3	Footwear	Sneakers	145

Key Finding: Top products include **Jewelry (171)**, **Sunglasses (161)**, and **Belt (161)** in Accessories; **Blouse (171)**, **Pants (171)**, and **Shirt (169)** in Clothing; and **Sandals (160)**, **Shoes (150)**, and **Sneakers (145)** in Footwear.

Q9: Subscription Affinity Among Repeat Buyers (>5 Purchases)

```
SELECT subscription_status,
       COUNT(customer_id) AS repeat_buyers
FROM customer_behavior.customer_data
WHERE previous_purchases > 5
GROUP BY subscription_status;
```

subscription_status	repeat_buyers
Yes	958
No	2518

Key Finding: Out of 3,476 repeat buyers with >5 purchases, **2,518 (72.4%) are unsubscribed**, highlighting a massive conversion opportunity.

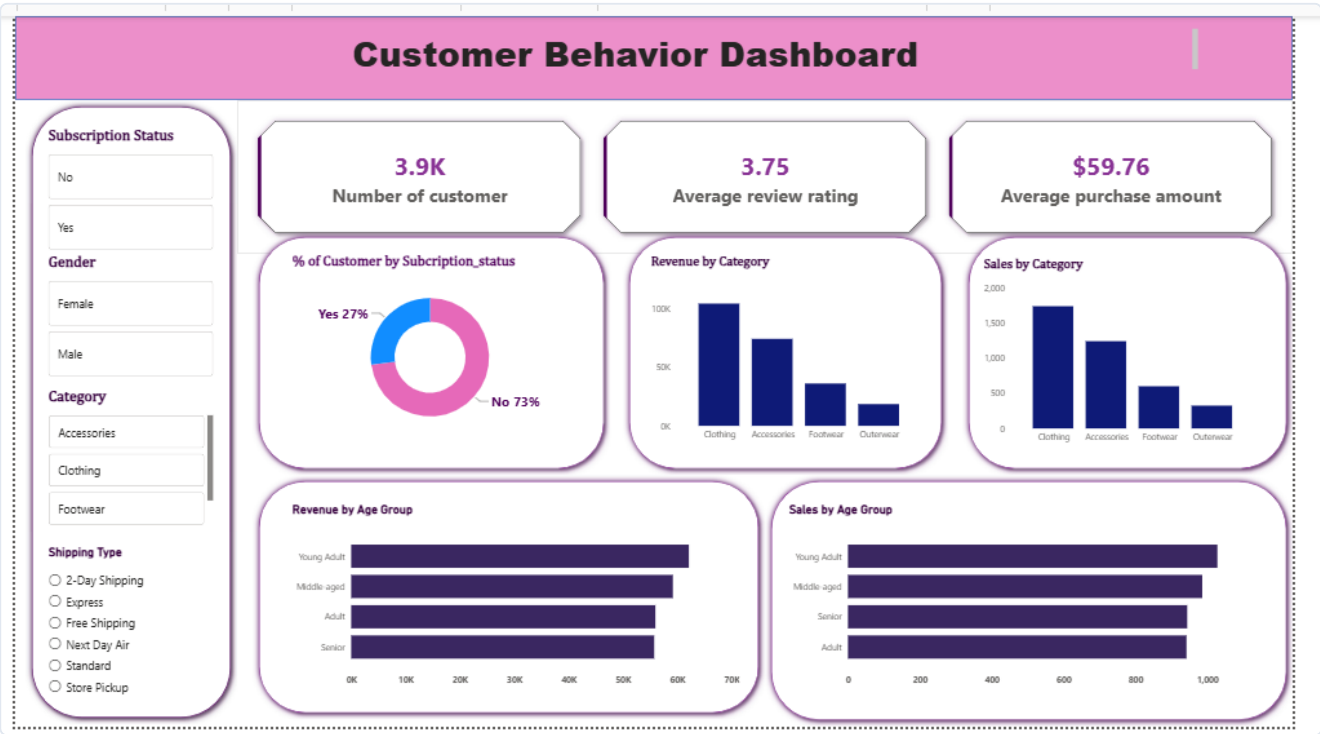
Q10: Revenue Contribution by Age Group

```
SELECT age_gab,
       SUM(purchase_amount) AS total_revenue
FROM customer_behavior.customer_data
GROUP BY age_gab
ORDER BY total_revenue DESC;
```

Key Finding: Revenue contribution is led by **Young Adults ($62,143)**, followed by **Middle-aged ($59,197)**, **Adults ($55,978)**, and **Seniors ($55,763)**.

age_gab	total_revenue
Young Adult	62143
Middle-aged	59197
Adult	55978
Senior	55763

3. Power BI Interactive Executive Dashboard



Dashboard Visual Components & Architecture

- **Executive KPI Header:** Cards displaying Total Customers (**3.9K**), Average Review Rating (**3.75**), and Average Purchase Amount (**$59.76**).
- **Subscription Penetration:** Donut chart showcasing that **73% of customers are non-subscribers** while **27% are subscribed**.
- **Category Performance Breakdown:** Column charts detailing **Revenue by Category** (Clothing leading with >$100K) alongside **Sales Volume by Category**.
- **Age Demographic Performance:** Horizontal bar charts comparing **Revenue by Age Group** and **Sales Count by Age Group** across Young Adult, Middle-aged, Senior, and Adult segments.
- **Dynamic Slicers Panel:** Interactive filtering by Subscription Status, Gender, Category, and Shipping Type (2-Day, Express, Free Shipping, Next Day Air, Standard, Store Pickup).

4. Strategic Business Recommendations

- **Target High-Frequency Unsubscribed Shoppers:** Since **72.4% of repeat buyers (>5 orders) are not subscribed**, introduce post-checkout loyalty prompts offering members-only shipping or early-access discounts.
- **Protect Profit Margins with Cart Thresholds:** High-value shoppers consistently spend $64–$97 even with discounts. Set a minimum cart threshold of **$65+** to prevent unnecessary margin erosion.
- **Category Inventory Allocation:** Maintain prioritized safety stock on top-selling SKUs (Blouses, Pants, Jewelry, Sandals) while monitoring promotional margins on discount-heavy lines like Hats and Sneakers.